

NAME: _____

TEACHER: _____

BAULKHAM HILLS HIGH SCHOOL

YEAR 12

HALF YEARLY EXAMINATION

2008

MATHEMATICS

EXTENSION 1

GENERAL INSTRUCTIONS:

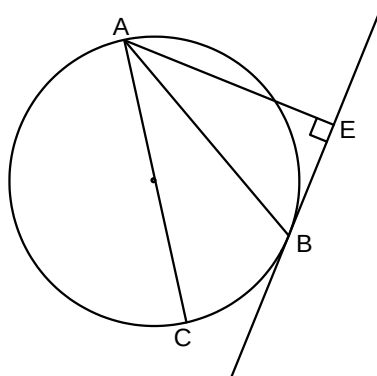
- Attempt **ALL** questions.
- Start each of the 7 questions on a new page.
- All necessary working should be shown.
- Write your teacher's name and your name on the cover sheet provided.
- At the end of the exam, staple your answers in order behind the cover sheet.
- Marks indicated for each question are only a guide and could change.

QUESTION 1**Marks**

- (a) If $P(x) = x^3 - 2x^2 + ax + 4$ is divisible by $(x + 2)$, what is the value of a ? **1**
- (b) Differentiate $\log_e x^2$. **2**
- (c) Sketch the graph $y = \frac{4}{x-1}$ **4**
Hence or otherwise solve $x + 2 < \frac{4}{x-1}$.
- (d) Find the solution for x if $\cos 2x = \cos x$ $0 \leq x \leq 2\pi$ **3**
- (e) Find $\int \sin^2 x \, dx$. **2**

QUESTION 2 (Start a new page)

- (a) The equation $3x^3 - 4x^2 + 2x + 1 = 0$ has roots α, β and γ . **3**
Find:
- (i) $2\alpha + 2\beta + 2\gamma$
- (ii) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$
- (b) Evaluate $\int_0^{\frac{\pi}{3}} \tan^2 x \, dx$. (Leave answer in exact form) **3**
- (c) Two points A and B are on the circumference of a circle and AC is the **3**
diameter. AE is perpendicular to the tangent at B . Prove AB bisects $\angle CAE$.



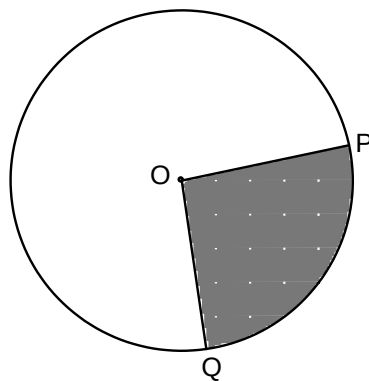
- (d) The graphs of $y = e^x$ and $y = e^{-2x}$ intersect at $x = 0$. **3**
Find the size of the acute angle between the curves at $x = 0$.

QUESTION 3 (Start a new page)**Marks**

- (a) The volume of a sphere is increasing at a rate of $5\text{cm}^3/\text{s}$.
At what rate is the surface area increasing when the radius is 20cm . 3
- (b) Show that the point $A(\frac{1}{2}, 4)$ lies on the line joining the points
 $P(-3, -3)$ and $Q(1, 5)$, and find the ratio in which it divides the line segment PQ . 3
- (c) Evaluate $\lim_{x \rightarrow 0} \frac{5x \cos 2x}{\sin x}$ showing your reasoning. 2
- (d) Use the method of mathematical induction to show that the expression
 $9^n - 8n - 1$ is divisible by 64 for all integers $n \geq 2$. 4

QUESTION 4 (Start a new page)

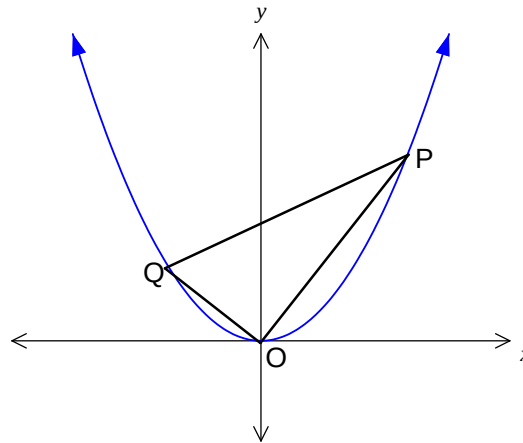
- (a) Find the area of sector POQ in the circle of radius 6cm and $\angle POQ = \frac{\pi}{3}$ 2



- (b) (ii) Express $\cos 2x - \sin 2x$ in the form $R \cos(2x + \alpha)$
where α is acute and $R > 0$. 2
- (ii) Sketch the curve $y = \cos 2x - \sin 2x$ for $0 \leq x \leq 2\pi$ 2
- (iii) Hence or otherwise solve:
 $\cos 2x - \sin 2x = 1$ for $0 \leq x \leq 2\pi$. 2

QUESTION 4 (Continued)
(c)

Marks



$P(2ap, ap^2)$ and $Q(2aq, aq^2)$ are points on the parabola $x^2 = 4ay$. It is given that $\angle POQ = 90^\circ$. O is the origin.

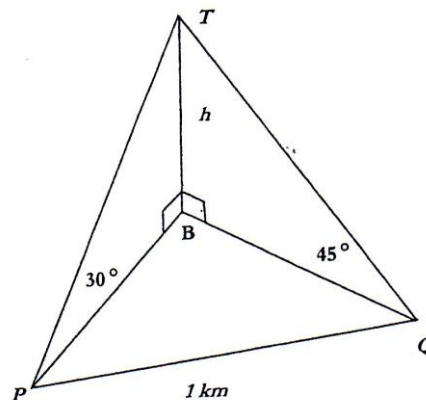
- (i) Prove that $pq = -4$. 1
- (ii) Find the co-ordinates of M , the midpoint of PQ . 1
- (iii) Prove that the Cartesian equation of the locus of M is $2ay = x^2 + 8a^2$. 2

QUESTION 5 (Start a new page)

Marks

- (a) Evaluate $\int_0^{\ln 4} 3e^x dx$ 2

(b)



The angle of elevation from a boat at P to a point T at the top of a vertical cliff is 30° . The boat sails 1 km to a second point Q , from which the angle of elevation of T is 45° . B is the point at the base of the cliff directly below T and h is the height of the cliff in metres. The bearings of B from P and Q are 50° and 290° respectively.

- (i) Show that $\angle PBQ = 120^\circ$. 1
- (ii) By finding expressions for PB and QB in terms of h , show that:

$$h = \frac{1000}{\sqrt{4 + \sqrt{3}}} \quad 4$$

QUESTION 5 (Continued)**Marks**

- (c) A body cools according to the equation:

$$\frac{dT}{dt} = -k(T - S)$$

where T is the temperature of the body at time t ,
 S is the temperature of the surroundings and
 k is a constant.

- (i) Show that $T = S + Ae^{-kt}$ satisfies the equation, where A is a constant. **1**

- (ii) A metal rod has an initial temperature of 470°C and cools to 250°C in 10 minutes.

The surrounding temperature is 30°C . **2**

(α) Find the value of A and show that $k = \frac{1}{10} \log_e 2$.

(β) Find how much longer it will take the rod to cool to 70°C , giving your answer to the nearest minute. **2**

QUESTION 6 (Start a new page)**Marks**

- (a) (i) Find the polynomial $P(x)$, if $P(x)$ has: **3**

- degree 4.
- factors $(x + 3)^2$ and $(x - 3)^2$ and
- a remainder of -50 when divided by $(x + 2)$.

- (ii) Sketch this curve.

- (b) An arc of a circle subtends an angle θ radians at the centre. The length of the arc is ℓ and the length of the chord is d . **3**

If $\ell : d = 4 : 3$, show that $3\theta - 8\sin \frac{\theta}{2} = 0$

- (c) Find the exact value of x if $\log_e(2\log_e x) = 1$ **2**

- (d) Find $\int 7^x dx$ **1**

- (e) An amount of \$25,000 is borrowed and an interest rate of 6% pa is charged. An amount M is repaid in monthly amounts.

- (i) Show that the amount owing after n months is: **2**

$$A_n = 25000(1.005)^n - M \left(\frac{1.005^n - 1}{0.005} \right)$$

- (ii) If the loan is paid off in 5 years, calculate the repayment M to the nearest dollar. **1**

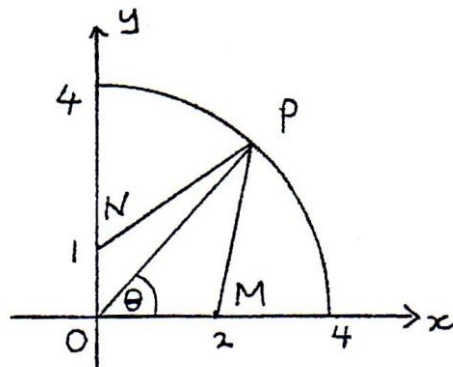
- (a) A particle is moving in a straight line and its velocity v metres/second at time t seconds is given by:

$$v = \frac{dx}{dt} = 1 - 2\sin 2t \quad t \geq 0$$

Initially the particle is at the origin.

- | | | |
|-------|---|---|
| (i) | Express the displacement x , as a function of t . | 2 |
| (ii) | Find the position of the particle when $t = \frac{\pi}{6}$. | 1 |
| (iii) | Find an expression for the acceleration $a = \frac{d^2x}{dt^2}$. | 1 |
| (iv) | Sketch the graph of the acceleration as a function of time, $0 \leq t \leq \pi$. | 1 |
| (v) | What is the maximum acceleration of the particle? | 1 |

(b)



The diagram shows the part of the circle $x^2 + y^2 = 16$ that lies in the first quadrant. The point $P(x, y)$ is on the circle with centre at the origin O . M is on the x -axis at $x = 2$ and N is on the y -axis at $y = 1$. The size of angle MOP is θ radians.

- | | | |
|-------|--|---|
| (i) | Show that the area, A of the quadrilateral $OMPN$ is given by
$A = 4\sin\theta + 2\cos\theta$ | 2 |
| (ii) | Find the value of $\tan\theta$ for which A is maximum. | 2 |
| (iii) | Hence determine in surd form the coordinates of P for which A is a maximum. | 2 |

Quest 1.

a) $P(x) = x^3 - 2x^2 + ax + 4$

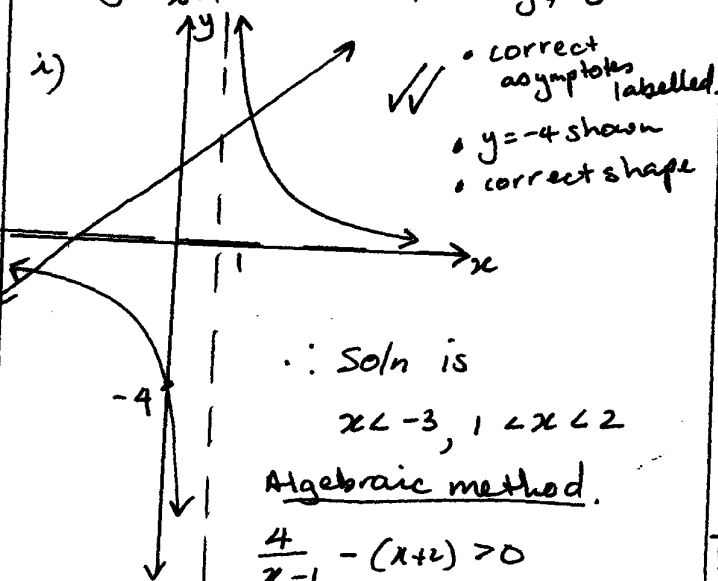
$P(-2) = -8 - 8 + 2a + 4 = 0$

$2a = -12$

$a = -6$ ✓

c) $y = \frac{4}{x-1}$ $x \neq 1$ $y \neq 0$

i)



∴ Soln is

$x < -3, 1 < x < 2$

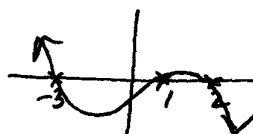
Algebraic method.

$\frac{4}{x-1} - (x+2) > 0$

$\frac{4 - (x-1)(x+2)}{x-1} > 0$

$(x-1)(6 - x - x^2) > 0$

$(x-1)(3+x)(2-x) > 0$



∴ $x < -3, 1 < x < 2$ ✓

b) $\frac{d}{dx} \log_e x^x = \frac{2x}{x^x}$ ✓
 $= \frac{2}{x}$ ✓

d) $\cos 2x = \cos x$ $0 \leq x \leq 2\pi$
 $2 \cos^2 x - 1 - \cos x = 0$ ✓

$(2 \cos x + 1)(\cos x - 1) = 0$

$\cos x = -\frac{1}{2}$ $\cos x = 1$ ✓

$x = \pi + \frac{\pi}{3}, \pi - \frac{\pi}{3}$ $x = 0, 2\pi$

∴ $x = 0, \frac{2\pi}{3}, \frac{4\pi}{3}, 2\pi$ ✓

(3)

e) $\int \sin^2 x \, dx = \int \frac{1}{2}(1 - \cos 2x) \, dx$ ✓
 $= \frac{1}{2}(x - \frac{1}{2} \sin 2x) + C$ ✓
 (2)

Question 2.

a) $3x^3 - 4x^2 + 2x + 1 = 0$

$\alpha + \beta + \gamma = -\frac{b}{a}$ $\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a}$
 $= \frac{4}{3}$ $= \frac{2}{3}$

$\alpha\beta\gamma = -\frac{d}{a}$
 $= -\frac{1}{3}$

∴ i) $2(\alpha + \beta + \gamma) = 2 \times \frac{4}{3}$

ii) $\frac{\alpha\beta + \alpha\gamma + \beta\gamma}{\alpha\beta\gamma} = \frac{\frac{2}{3}}{-\frac{1}{3}} = -2$ ✓
 (3)

b) $\int_0^{\frac{\pi}{3}} \tan^2 x \, dx = \int_0^{\frac{\pi}{3}} (\sec^2 x - 1) \, dx$ ✓
 $= [\tan x - x]_0^{\frac{\pi}{3}}$ ✓
 $= (\tan \frac{\pi}{3} - \frac{\pi}{3}) - 0$
 $= \sqrt{3} - \frac{\pi}{3}$ ✓ or $\frac{3\sqrt{3} - \pi}{3}$ (3)

c) Construct CB

Proof: $\hat{ABE} = \hat{ACB}$ (angle formed between
 ✓ = x chord and tangent at
 pt of contact is equal
 to the angle in the
 alternate segment)

✓ $\hat{ABC} = 90^\circ$ (angle in a semi circle)

∴ $\hat{EAB} = 180 - (90 + x)$ (angle sum of Δ)
 $= 90 - x$

✓ $\hat{CAB} = 90 - x$ (angle sum of Δ)
 $= \hat{EAB}$

∴ AB bisects $\hat{CAE}$

(3)

quest 2 cont.

d) $y = e^x$
 $\frac{dy}{dx} = e^x$
 $m_1 = e^0 = 1$

$y = e^{-2x}$
 $\frac{dy}{dx} = -2e^{-2x}$
 $m_2 = -2 \cdot e^0 = -2$

$$\therefore \tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$= \left| \frac{1 - (-2)}{1 + (-2)} \right| = 3$$

$$\theta = 71^\circ 34' \quad \checkmark \quad (3)$$

Question 3

a) $V = \frac{4}{3}\pi r^3$ $SA = 4\pi r^2$ $r = 20$

$$\frac{dV}{dt} = 5$$

$$\frac{dV}{dr} = 4\pi r^2$$

$$\frac{dA}{dr} = 8\pi r$$

$$\frac{dV}{dt} = \frac{dV}{dr} \cdot \frac{dr}{dt}$$

$$5 = 4\pi r^2 \times \frac{dr}{dt}$$

$$\frac{dr}{dt} = \frac{5}{4\pi r^2}$$

$$\frac{dA}{dt} = \frac{dA}{dr} \cdot \frac{dr}{dt}$$

$$= 8\pi r \cdot \frac{5}{4\pi r^2}$$

$$= \frac{10}{r}$$

at $r = 20$

$$\frac{dA}{dt} = \frac{10}{20} = 0.5$$

at $0.5 \text{ m}^2/\text{sec}$ A increasing $\checkmark$ (3)

b) $P(-3, -3)$ $Q(1, 5)$

$$m_{PQ} = \frac{5 - (-3)}{1 - (-3)} = \frac{8}{4} = 2$$

line PQ:

$$y + 3 = 2(x + 3)$$

$$y = 2x + 3$$

does $A(\frac{1}{2}, 4)$ satisfy line $\checkmark$

ie. $4 = 2 \times \frac{1}{2} + 3$

$= 4$ True

ratio

$$x = \frac{m x_2 + n x_1}{m + n}$$

$$\frac{1}{2} = \frac{m \times 1 + n \times (-3)}{m + n}$$

$m + n = 2m - 6n$

$7n = m$ $\therefore \frac{m}{n} = \frac{7}{1}$ or $m:n = 7:1$ $\checkmark$ (3)

c) $\lim_{x \rightarrow 0} \cos 2x \cdot \frac{5x}{\sin x} = \lim_{x \rightarrow 0} \cos 2x \cdot 5 \lim_{x \rightarrow 0} \frac{x}{\sin x}$

$= 5 \checkmark$ $\checkmark$ - some working (2)

d) $9^n - 8n - 1$

let $n = 2$

$$9^2 - 16 - 1 = 81 - 17 = 64$$

which is divisible by 64 $\checkmark$

assume true ... true for $n = 2$

for $n = k$

$$\therefore \frac{9^k - 8k - 1}{64} = P \text{ (some integer)}$$

prove true for $n = k + 1$

ie. $9^{k+1} - 8(k+1) - 1$

$$= 9^k \times 9 - 8k - 9$$

[but $9^k = 64P + 8k + 1$]

$$= (64P + 8k + 1) \times 9 - 8k - 9$$

$$= 9 \times 64P + 72k + 9 - 8k - 9$$

$$= 64(9P + k) \text{ which is divisible by 64} \quad \checkmark$$

$\therefore$ if true for $n = k$ then true for $n = k + 1$

Since true for $n = 2$ then by M.I. also true for $n = 2 + 1 = 3$, $n = 3 + 1 = 4$ for all integers $n \geq 2$. $\checkmark$ (4)

Question 4

a) $A = \frac{1}{2} r^2 \theta$

$$= \frac{1}{2} \times 36 \times \frac{\pi}{3}$$

$$= 6\pi \text{ m}^2 \quad \checkmark$$

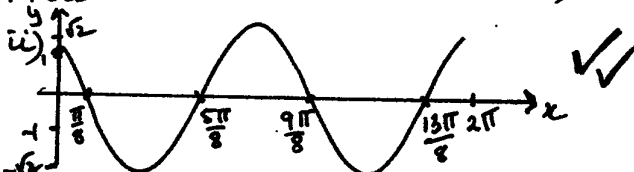
(2)

b) $\cos 2x - \sin 2x = \cos 2x \cos \alpha - \sin 2x \sin \alpha$

$\therefore \tan \alpha = 1$ $\checkmark$ $R = \sqrt{2}$

$\alpha = \frac{\pi}{4}$

$\therefore \cos 2x - \sin 2x = \sqrt{2} \cos(2x + \frac{\pi}{4})$ $\checkmark$



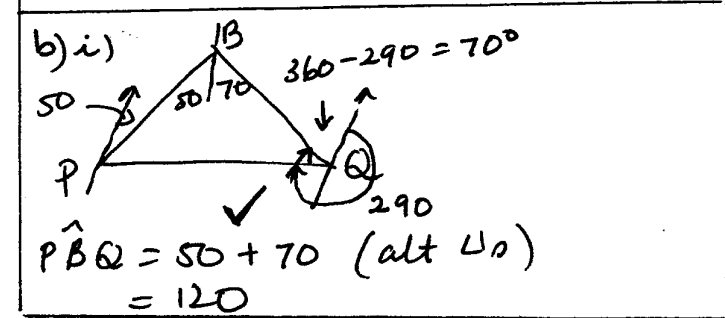
quest 4 cont.

ii) $\cos 2x - \sin 2x = 1$ $0 \leq x \leq 2\pi$
 $\sqrt{2} \cos(2x + \frac{\pi}{4}) = 1$ $\frac{\pi}{4} \leq 2x + \frac{\pi}{4} \leq \frac{7\pi}{4}$
 $\cos(2x + \frac{\pi}{4}) = \frac{1}{\sqrt{2}}$
 $2x + \frac{\pi}{4} = \frac{\pi}{4}, \frac{7\pi}{4}, \frac{9\pi}{4}, \frac{15\pi}{4}, \frac{17\pi}{4}$
 $2x = 0, \frac{6\pi}{4}, \frac{8\pi}{4}, \frac{14\pi}{4}, \frac{16\pi}{4}$
 $x = 0, \frac{3\pi}{4}, \pi, \frac{7\pi}{4}, 2\pi$

c) i) $M_{PO} = \frac{ap^2 - 0}{2ap - 0} = \frac{p}{2}$ $M_{QO} = \frac{q}{2}$
 Since $M_{PO} \times M_{QO} = -1$ $\angle OPQ = 90^\circ$
 $\therefore \frac{p}{2} \times \frac{q}{2} = -1$
 $pq = -4$

ii) mid pt PQ
 $x = \frac{2ap + 2aq}{2}$ $y = \frac{ap^2 + aq^2}{2}$
 $= a(p+q)$ $= \frac{a}{2}(p^2 + q^2)$
 $\therefore M = (a(p+q), \frac{a}{2}(p^2 + q^2))$
 iii) now $\frac{x}{a} = p+q$ $y = \frac{a}{2}((p+q)^2 - 2pq)$
 $= \frac{a}{2}((\frac{x}{a})^2 - 2x - 4)$
 $= \frac{a}{2}(\frac{x^2}{a^2} + 8)$
 $= \frac{x^2}{2a} + 4a$
 $2ay = x^2 + 8a^2$ (4)

Questions
 a) $\int_0^{\ln 4} 3e^x dx = [3e^x]_0^{\ln 4}$
 $= 3e^{\ln 4} - 3e^0$
 $= 3 \times 4 - 3$
 $= 9$ (2)



ii) In $\triangle TBP$ $\tan 30 = \frac{h}{PB}$
 $PB = \frac{h}{\tan 30} = \sqrt{3}h$
 In $\triangle TBQ$ $\tan 45 = \frac{h}{BQ}$
 $BQ = \frac{h}{\tan 45} = h$
 $\therefore$ in $\triangle PBQ$
 $1000m^2 = PB^2 + BQ^2 - 2PB \cdot BQ \cdot \cos 120$
 $= (\sqrt{3}h)^2 + h^2 - 2(\sqrt{3}h \cdot h) \cdot (-\frac{1}{2})$
 $1000 = 3h^2 + h^2 + \sqrt{3}h^2$
 $= h^2(4 + \sqrt{3})$
 $h^2 = \frac{1000}{4 + \sqrt{3}}$
 $h = \frac{10\sqrt{10}}{\sqrt{4 + \sqrt{3}}}$ (5)

c) i) $T = S + Ae^{-kt}$
 then $\frac{dT}{dt} = -k \cdot Ae^{-kt}$
 but $Ae^{-kt} = T - S$
 $\therefore \frac{dT}{dt} = -k(T - S)$ as required
 ii) $t=0, T=470$ $t=10, T=250$
 $S=30$
 $\therefore T = 30 + Ae^{-kt}$
 $470 = 30 + Ae^0$
 $A = 440$
 $T = 30 + 440e^{-10k}$
 $250 = 30 + 440e^{-10k}$
 $\frac{220}{440} = e^{-10k}$
 $\ln 0.5 = -10k \ln e$
 $k = -\frac{\ln 0.5}{10}$
 $= -\frac{(-\ln 2)}{10}$
 $k = \frac{1}{10} \ln 2$

(b) $70 = 30 + 440e^{-kt}$
 $\frac{40}{440} = e^{-kt}$
 $\ln \frac{1}{11} = -\frac{1}{10} \ln 2 \cdot t$
 $t = 34.59$
 $\approx 35 \text{ min}$ (5)

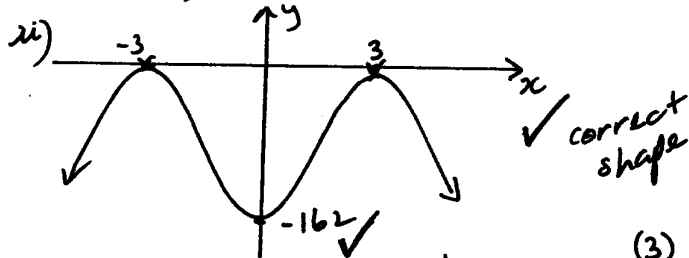
Question 6.

a) i) $p(x) = a(x+3)^2(x-3)^2$ /12.

$$p(-2) = a(1)(-5)^2 = -50$$

$$\therefore a = -2$$

$$\therefore p(x) = -2(x+3)^2(x-3)^2$$



(3)

b)

$$\sin \frac{\theta}{2} = \frac{d/2}{r}$$

$$r \sin \frac{\theta}{2} = \frac{d}{2}$$

$$\Rightarrow d = 2r \sin \frac{\theta}{2}$$

now $\Rightarrow l = r\theta$

$$l : d = r\theta : 2r \sin \frac{\theta}{2} = 4 : 3$$

$$\frac{r\theta}{2r \sin \frac{\theta}{2}} = \frac{4}{3}$$

$$3\theta = 8 \sin \frac{\theta}{2}$$

$$\text{or } 3\theta - 8 \sin \frac{\theta}{2} = 0 \quad (3)$$

c) $\log_e(2 \log_e x) = 1$

$$e = 2 \log_e x$$

$$\frac{e}{2} = \log_e x$$

$$\therefore e^{\frac{e}{2}} = x$$

(2)

d) $\int 7^x dx = \frac{1}{\ln 7} \cdot 7^x + C$ ✓
must have

(1)

e) $r = \frac{0.06}{12} = 0.005$

$$A_1 = 25000(1.005) - m$$

$$A_2 = A_1(1.005) - m$$

$$= 25000(1.005)^2 - m(1.005) - m$$

$$A_3 = A_2(1.005) - m$$

$$= 25000(1.005)^3 - m(1.005)^2 - m(1.005) - m$$

$$\therefore A_n = 25000(1.005)^n - m(1.005)^{n-1} - m(1.005)^{n-2} - \dots$$

$$= 25000(1.005)^n - m(1 + 1.005 + 1.005^2 + \dots)$$

$$= 25000(1.005)^n - m \left(\frac{1.005^n - 1}{0.005} \right)$$

ii) Syrs = 60 for n

$$M = 25000(1.005)^{60} \times \frac{0.005}{1.005^{60} - 1}$$

$$= \$483.32$$

$$\approx \$483$$

(3)

Question 7.

/12

a) $V = \frac{dx}{dt} = 1 - 2\sin 2t$

i) $x = t + \cos 2t + C$ ✓

at $t=0$ $x=0$

$$0 = 0 + \cos 0 + C$$

$$C = -1$$

$$\therefore x = t + \cos 2t - 1$$

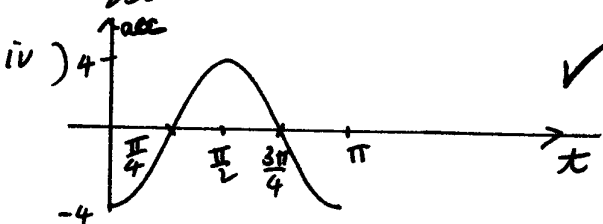
ii) $t = \frac{\pi}{6}$

$$x = \frac{\pi}{6} + \cos \frac{\pi}{3} - 1$$

$$= \left(\frac{\pi}{6} - \frac{1}{2} \right) \text{m or } 0.02 \text{m to the right}$$

iii) $\frac{dx}{dt} = 1 - 2\sin 2t$

$$\frac{d^2x}{dt^2} = -4 \cos 2t$$



v) max accel. = 4 m/sec^2 ✓

(6)

Quest 7. Cont.

b. Area of OMPN

$$= A_{OPM} + A_{OPN}$$

$$\therefore A_{OPM} = \frac{1}{2} \times 4 \times 2 \times \sin \theta \quad \checkmark$$
$$= 4 \sin \theta$$

$$A_{OPN} = \frac{1}{2} \times 1 \times 4 \times \sin(90 - \theta) \quad \checkmark$$
$$= 2 \cos \theta$$

$$\therefore \text{Total Area} = 4 \sin \theta + 2 \cos \theta$$

$$\text{ii) } \frac{dA}{d\theta} = 4 \cos \theta - 2 \sin \theta = 0$$

$$4 \cos \theta = 2 \sin \theta$$

$$2 = \tan \theta \quad \checkmark$$

$$\frac{d^2 A}{d\theta^2} = -4 \sin \theta - 2 \cos \theta$$

$$= -(4 \sin \theta + 2 \cos \theta)$$

$$\therefore \frac{d^2 A}{d\theta^2} < 0 \text{ for any value of } \theta \quad \checkmark$$

$$\text{of } \theta \quad (-4.47 \text{ when using } \theta = 63.4^\circ)$$

$$\therefore \text{for A maximum } \tan \theta = 2$$

iii) let P be (x, y)

$$x^2 + y^2 = 16$$

$$\tan \theta = \frac{y}{x} = 2 \quad \checkmark$$

$$y = 2x$$

$$\therefore x^2 + 4x^2 = 16$$

$$5x^2 = 16$$

$$x = \frac{4}{\sqrt{5}}$$

$$y = 2 \times \frac{4}{\sqrt{5}} = \frac{8}{\sqrt{5}}$$

$$\therefore P \text{ is } \left(\frac{4}{\sqrt{5}}, \frac{8}{\sqrt{5}} \right) \quad \checkmark$$

(b)